Predicting Wine Cultivar with Logistic Regression

Abstract. We classify wines by cultivar using logistic regression on the wine dataset.

Data. The wine dataset has 178 samples and 13 numeric chemical features (alcohol, malic acid, ash, magnesium, flavanoids, color intensity, proline, and others). The target variable is the cultivar class. Features are standardized before modeling.

Method. We fit a multinomial logistic regression with a 75/25 train/test split and report accuracy on the held-out set. The independent variables are the 13 measurements.

Results. The model reaches about 0.97 accuracy. Flavanoids and proline are the most informative features. We conclude logistic regression is a strong baseline for this task.